ECON 7201
Applied Econometrics
Assignment 2

Due Date

```
install.packages("tikzDevice")
library(tikzDevice)
```

November 5, 2026 at 11:59 PM

Directions

Answer all questions. Upload both a PDF and Quarto file to your website in a new blog post.

Git and GitHub

1. (a) Create a new R project in your **econ_3201** directory called **assignment_2**.
   (b) Download the assignment PDF and Quarto file the **assignment_2** folder.
   (c) Commit and push the changes to your **econ_3201** repository on GitHub.com.

LaTeX

Matrices are created in LaTeX using the `\begin{bmatrix}...\end{bmatrix}` command. To separate entries along the same row, use `&`. To end a line, use `\\`. To make vertical elipses (:), use `\vdots`. Practice writing the following matrices and vectors in LaTeX. Write the following matrices in LaTeX.

2. (a)

Tip

$$X'X = \begin{bmatrix} n & \sum_{i=1}^{n} x_{1i} & \sum_{i=1}^{n} x_{2i} \\ \sum_{i=1}^{n} x_{1i} & \sum_{i=1}^{n} x_{1i}^2 & \sum_{i=1}^{n} x_{1i}x_{2i} \\ \sum_{i=1}^{n} x_{2i} & \sum_{i=1}^{n} x_{1i}x_{2i} & \sum_{i=1}^{n} x_{2i}^2 \end{bmatrix}$$

(b)

Tip

$$\Omega = \begin{bmatrix} \sigma_1^2 & 0 & 0 & 0 \\ 0 & \sigma_2^2 & 0 & 0 \\ 0 & 0 & \sigma_3^2 & 0 \\ 0 & 0 & 0 & \sigma_4^2 \end{bmatrix}$$

R

3. In this question we compare standard errors based on (incorrect) asymptotic assumptions with those based on alternate (appropriate) estimator (White). Consider one sample drawn from the following data generating process (DGP) which we will simulate in R:

```
set.seed(123)
n <- 25
x <- rnorm(n,mean=0.0,sd=1.0)
beta0 <- 1
beta1 <- 0
## x is irrelevant in this model, the data generating process is as follows:
dgp <- beta0 + beta1*x
## The residual is heteroskedastic by construction
e <- x^2*rnorm(n,mean=0.0,sd=1.0)
y <- dgp + e
```

(a) Compute the OLS estimator of β_2 and its standard error using the `lm()` command in R for the model $y_i = \beta_1 + \beta_2 x_i + \epsilon_i$ based on the DGP given above.

Tip

$$y_i = \beta_1 + \beta_2 x_i + \epsilon_i$$

```
estimator <- lm(y~x) #This is software regression we are getting the computer to do it
summary(estimator)
```

(b) Next, compute the standard error of $\hat{\beta}_2$ by computing $\hat{\sigma}^2(X'X)^{-1}$ in R using matrix commands, and verify that the two standard error estimates are identical.

Tip

```
#Create X matrix
X <- cbind(1,x)
beta_hat <- solve(t(X) %*% X) %*% t(X) %*% y
```

$$\hat{\beta} = (X'X)^{-1}X'y$$

```
#Residuals
e <- y - X %*% beta_hat
```

$$\sigma^2 = \frac{\sum e^2}{n-2}$$

```
error_var <- sum(e^2)/(n-2)
```

Variance-Covariance Matrix

$$\hat{Var}\hat{\beta} = \sigma^2(X'X)^{-1}$$

```
var_beta <- error_var *solve(t(X) %*% X)
```

Standard Errors

```
se <- sqrt(diag(var_cov))
```

```
se_lm <- summary(estimator)$coefficients[2,2]
```

ensure errors are equal

```
all.equal(se_lm, se)
```

(c) Compute White's heteroskedasticity consistent covariance matrix estimator using matrices in R and report the White estimator of the standard error of $\hat{\beta}_2$. Compare this with that from 3 (a) above.

Tip

$$\hat{Var}_{white}(\beta) = (X'X)^{-1}X'\hat{\Omega}X(X'X)^{-1}$$

```
e
e_squared <- e^2
#Omega Matrix
Omega <- diag(as.vector(e)^2)
#Solve inverse of transpose * X
XtX_inv <- solve(t(X) %*% X)
#input into formula
white_vcov <- XtX_inv %*% t(X) %*% Omega %*% X %*% XtX_inv
#White standard error
white_se <- sqrt(diag(white_vcov))[2]
se_lm
white_se
```

4. Let $\hat{\theta}$ be an estimator for the population parameter θ. $\hat{\theta}$ is said to be unbiased if $E(\hat{\theta}) = \theta$. That is, if the mean of the sampling distribution of $\hat{\theta}$ is equal to the true population value.

   Consider the model

   $$y_i = \beta_0 + \beta_1 x_{1,i} - \beta_2 x_{2,i} + \epsilon_i.$$

   Lets provide empirical evidence that the ordinary least squares estimators $\hat{\beta}_0$, $\hat{\beta}_1$, and $\hat{\beta}_2$ are unbiased estimators of β_0, β_1, β_2, respectively, using R.

   (a) Set the seed to 1, i.e., `set.seed(1)`.
   (b) Set the number of observations $n = 100$
   (c) Generate the following model

   $$y_i = 2 + 3.5x_{1,i} - 9.2x_{2,i} + \epsilon_i,$$

   where $x_1 \sim N(3, 6)$, $x_2 \sim N(2, 4)$, and $\epsilon \sim N(0, 100)$. To create a normally distributed variable, use the `rnorm(n, mean, sd)` command in R.
   (d) Estimate the model coefficients using the `lm()` command. (Search `?lm()` in the console for more info).

(e) Using a `for()` loop, replicate the model above $M = 1000$ times and save the coefficient estimates from each iteration.
(f) Using `hist()`, plot the sampling distributions of the coefficient estimates, β_1 and β_2.
(g) Add a vertical line to each figure at the mean of the respective variable. Search `?abline()` in your console.

Tip

```r
#part a
set.seed(1)

#part b
n <- 100

#part c
x_1 <- rnorm(n,3,6)
x_2 <- rnorm(n,2,4)
epsilon <- rnorm(n,0,100)

Y <- 2 + 3.5*x_1 + 9.2*x_2 + epsilon

#part d
OLS <- lm(Y~x_1+x_2)
summary(OLS)

#part e
M <- 1000
## Storing Coefficient as vectors
beta0_hat <- numeric(M)
beta1_hat <- numeric(M)
beta2_hat <- numeric(M)

for (i in 1:M) {

  x1 <- rnorm(n, mean = 3, sd = 6)
  x2 <- rnorm(n, mean = 2, sd = 4)

  e <- rnorm(n, mean = 0, sd = 100)

  y <- 2 + 3.5*x1 - 9.2*x2 + e

  model <- lm(y ~ x1 + x2)

  beta0_hat[i] <- coef(model)[1]
  beta1_hat[i] <- coef(model)[2]
  beta2_hat[i] <- -coef(model)[3]

}

#part f & g
## Beta1 Histogram
hist(beta1_hat,
     main = "Sampling Distribution of Beta 1",
     xlab = "Beta 1 Estimates",
     col = "lightblue")
abline(v = mean(beta1_hat), col = "pink", lwd = 2)

## Beta2 Histogram
hist(beta2_hat,
     main = "Sampling Distribution of Beta 2",
```